

Dynamic Orientation Estimation with Sensor Fusion Of 9 - Axis Inertial Measurement Unit (IMU) Sensors

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1. Introduction

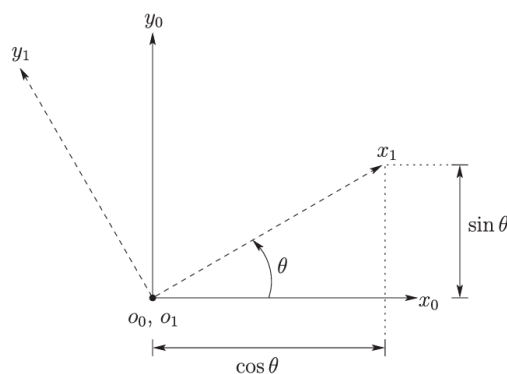
An Inertial Measurement Unit (IMU) is an electronic device used to estimate the orientation and motion of a rigid body. A typical 9-axis IMU consists of a three-axis accelerometer, three-axis gyroscope, and three-axis magnetometer. The accelerometer measures the total acceleration acting on the sensor, including gravity and any linear acceleration. The gyroscope measures angular velocity about the three body axes, while the magnetometer measures the Earth's magnetic field, providing a reference for heading.

Each sensor has inherent limitations when used independently. Integrating gyroscope measurements provides smooth short-term orientation estimates, but small sensor biases accumulate over time, causing drift. The accelerometer can estimate roll and pitch from the direction of gravity when the sensor is stationary or moving slowly, but its measurements become unreliable during linear motion. The magnetometer provides heading (yaw) information but is sensitive to magnetic disturbances caused by nearby ferromagnetic materials and electrical equipment. Sensor fusion combines the complementary strengths of these sensors to obtain a more accurate and robust orientation estimate than any individual sensor alone. Before developing the sensor fusion algorithm, we first introduce the mathematical representation of three-dimensional rotations using rotation matrices and Euler angles.

2. Representing Rotations

In order to represent the relative position and orientation of one rigid body with respect to another, we will rigidly attach coordinate frames to each body, and then specify the geometric relationships between these coordinate frames.

Rotations In 2D



Symbol	Meaning
R_B^E	Rotation matrix from Earth to Body frame
p^E	Point p Expressed in Earth frame
p^B	Point p Expressed in Body frame

Two coordinate frames shown, with frame $o_1x_1y_1$ obtained by rotating frame $o_0x_0y_0$ by an angle θ . When we specify the coordinate vectors for the axes of frame $o_1x_1y_1$ with respect to coordinate frame $o_0x_0y_0$:

$$R_1^0 = \begin{pmatrix} | & | \\ x_1^0 & y_1^0 \\ | & | \end{pmatrix}$$

Where, x_1^0 and y_1^0 are the coordinates of the unit vectors x_1 and y_1 in the $o_0x_0y_0$ coordinate frame. Matrix R_1^0 is called the Rotation Matrix & for the 2D case, it can be calculated as:

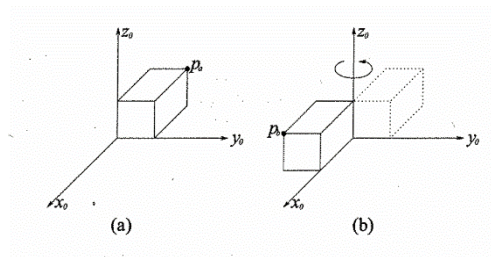
$$R_1^0 = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$$

NOTE: A rotation matrix may represent 3 operations, let $p = [p_1 \ p_2 \ p_3]$ represent a point in 3D space or a vector.

I. Changing the frame (change coordinates / change basis) of a vector or another frame

E.g: $p^a = R_b^a p^b$ will change the basis of vector p from frame b to frame a

II. Representing a rotation in the current frame.



Notice the rotation of point p on the box. Suppose, there is a frame (x_1, y_1, z_1) : **frame 1**, attached to the box, in (a) (x_1, y_1, z_1) coincides with (x_0, y_0, z_0) : **frame 0**. In (b), when the box is rotated, (x_1, y_1, z_1) rotates with the body. For (b), We can calculate R_1^0 by writing basis vectors of frame 1, w.r.t frame 0. Then, $p^0 = R_1^0 p^1$ will transform the basis from frame 1 to frame 0. p^0 represents the rotated vector p according to **frame 0**.

III. Representing the orientation of one frame relative to another.

The rotation matrix carries the relative orientation of the two coordinate frames.

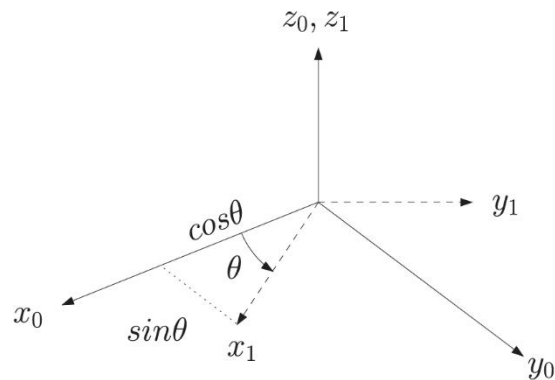
$$R_1^0 = \begin{pmatrix} | & | \\ x_1^0 & y_1^0 \\ | & | \end{pmatrix}$$

Question 1: Verify, $p^0 = R_1^0 p^1$ with $p = [1 \ 0]$ and $R_1^0 = \begin{bmatrix} \cos(45^\circ) & -\sin(45^\circ) \\ \sin(45^\circ) & \cos(45^\circ) \end{bmatrix}$

Question 2: Use the same rotation matrix to rotate p^1 in the current frame

Rotations In 3D

Rotation about z_0 by an angle θ .



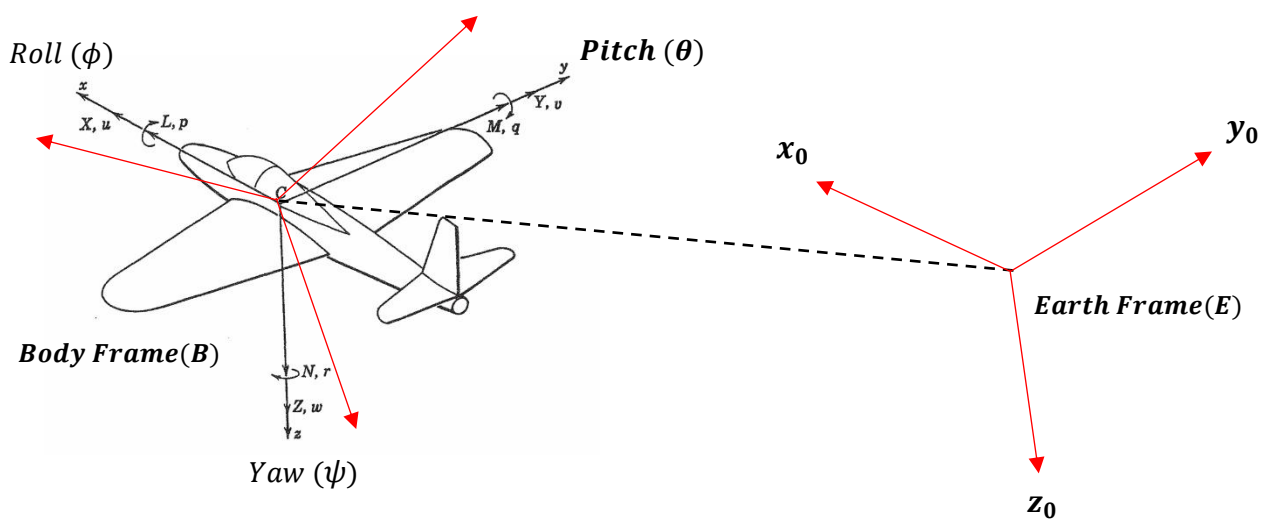
$$R_1^0 = \begin{bmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix} = R_{z,\theta}$$

Similarly, Rotations about x and y axes,

$$R_{x,\theta} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\theta) & -\sin(\theta) \\ 0 & \sin(\theta) & \cos(\theta) \end{bmatrix}$$

$$R_{y,\theta} = \begin{bmatrix} \cos(\theta) & 0 & \sin(\theta) \\ 0 & 1 & 0 \\ -\sin(\theta) & 0 & \cos(\theta) \end{bmatrix}$$

Euler Angles



The term Euler Angles refers to the angles of rotation (ψ, θ, ϕ) needed to go from one coordinate system to another using the specific sequence of rotations.

The most common sequence is:

$$\mathbf{Yaw}(\psi) \rightarrow \mathbf{Pitch}(\theta) \rightarrow \mathbf{Roll}(\phi)$$

$$\mathbf{Z} \rightarrow \mathbf{Y} \rightarrow \mathbf{X}$$

$$p^{Body} = [R_x(\phi)R_y(\theta)R_z(\psi)]p^{Earth}$$

$$p^{body} = R_E^B p^{Earth}$$

$$\therefore p^{Earth} = [R_x(\phi)R_y(\theta)R_z(\psi)]^{-1} p^{Body} = R_B^E p^{Body}$$

Gravity in the body frame can be written as,

$$g^B = R_E^B g^E = R_E^B \begin{bmatrix} 0 \\ 0 \\ g \end{bmatrix}$$

Since $g^E = \begin{bmatrix} 0 \\ 0 \\ g \end{bmatrix}$, what we need is the 3rd row of $R_x(\phi)R_y(\theta)R_z(\psi)$ to calculate g^B

$$\therefore g^B = \begin{bmatrix} g \sin(\theta) \\ -g \cos(\theta) \sin(\phi) \\ -g \cos(\theta) \cos(\phi) \end{bmatrix}$$

Note: Euler angles suffer from gimbal lock but are easy to visualize and are therefore used in this laboratory. Industrial navigation systems typically use quaternions instead.

3. Why Sensor Fusion?

- Overcome limitations of individual sensors (Noise, uncertainty, Bias, Drift, Sampling rates)
- Estimate quantities that are not directly measurable or too expensive to measure.

4. Attitude Estimation 6 DOF IMU Data (Accelerometer + Gyroscope)

4.1. Sensors And Sensor Models

- I. 3 – Axis Accelerometer: Measures linear acceleration along 3 principal axes x, y, z of the sensor (m/s^2)
- II. 3 – Axis Gyroscope: Measures angular velocity along 3 principal axes x, y, z of the sensor (rad/s)

Accelerometer

Measures accelerations along each of the body frame axes.

$$\hat{a} = a^B = \begin{bmatrix} a_x \\ a_y \\ a_z \end{bmatrix} = \underbrace{\frac{d\hat{v}}{dt}}_{\substack{\text{Linear} \\ \text{Acceleration}}} + \underbrace{\hat{\omega}_b \times \hat{v}}_{\substack{\text{Rotational} \\ \text{translation}}} - \underbrace{R_0^1 \begin{bmatrix} 0 \\ 0 \\ g \end{bmatrix}}_{\text{gravity}} + \underbrace{\beta(t)}_{\text{Bias}} + \underbrace{\eta(t)}_{\text{Noise}}$$

If the body is at rest,

$$a^B = \begin{bmatrix} a_x \\ a_y \\ a_z \end{bmatrix} = \begin{bmatrix} g \sin(\theta) \\ -g \cos(\theta) \sin(\phi) \\ -g \cos(\theta) \cos(\phi) \end{bmatrix} + \begin{bmatrix} | \\ \text{Noise} \\ | \end{bmatrix} + \begin{bmatrix} | \\ \text{Bias} \\ | \end{bmatrix}$$

Gyroscope

Measures angular velocity along each of the body axes.

$$\omega^B = \begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix} = \begin{bmatrix} p \\ q \\ r \end{bmatrix} = \begin{bmatrix} | \\ \omega_{true} \\ | \end{bmatrix} + \begin{bmatrix} | \\ \text{Bias} \\ | \end{bmatrix} + \begin{bmatrix} | \\ \text{Noise} \\ | \end{bmatrix}$$

Transforming body angular rates from Gyroscope to Euler Rates in the fixed frame,

$$\begin{bmatrix} \dot{\phi} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} 1 & \sin(\phi) \tan(\theta) & \cos(\phi) \tan(\theta) \\ 0 & \cos(\phi) & -\sin(\phi) \end{bmatrix} \begin{bmatrix} p \\ q \\ r \end{bmatrix}$$

4.2. System Model

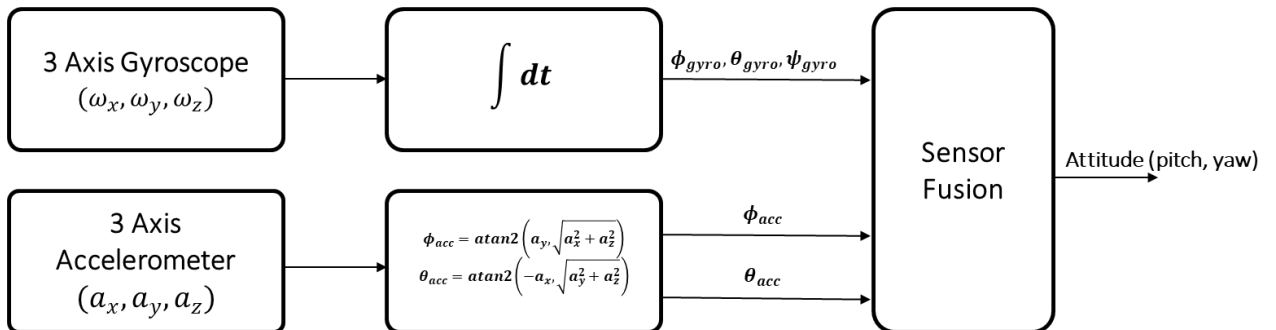


Figure 1: Overview of the System

Process model

Euler-rate kinematics: The gyro rates p, q, r (rad/s , *body* x, y, z) are mapped to Euler rates. Note that, accelerometer carries no heading information:

$$\begin{aligned}\dot{\phi} &= p + \sin(\phi) \tan(\theta) q + \cos(\phi) \tan(\theta) r \\ \dot{\theta} &= \cos(\phi) q - \sin(\phi) r\end{aligned}$$

State vector:

$$x = [\phi \quad b_\phi \quad \theta \quad b_\theta]^T$$

Where, $\phi = \text{roll}$, $\theta = \text{pitch}$, and b_ϕ, b_θ, b_ψ are *bias* states modeled as random-walks. The gyroscope bias changes slowly with temperature and time. Therefore, it is estimated online as part of the Kalman filter state.

Discrete - time process model.

$$\begin{aligned}\phi_k &= \phi_{k-1} + \Delta t(\dot{\phi} - b_\phi) \\ b_{\phi,k} &= b_{\phi,k-1} \\ \theta_k &= \theta_{k-1} + \Delta t(\dot{\theta} - b_\theta) \\ b_{\theta,k} &= b_{\theta,k-1}\end{aligned}$$

$$x_k = Ax_{k-1} + Bu_{k-1} + w_{k-1}, \quad w \sim N(0, Q)$$

$$A = \begin{bmatrix} 1 & -\Delta t & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -\Delta t \\ 0 & 0 & 0 & 1 \end{bmatrix}, B = \begin{bmatrix} \Delta t & 0 \\ 0 & 0 \\ 0 & \Delta t \\ 0 & 0 \end{bmatrix}, U = \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \end{bmatrix}, Q = 0.01I_4$$

Measurement model

At rest, the accelerometer measures only gravity. Therefore, the direction of the gravity vector in the body frame uniquely determines roll and pitch.

$$\begin{aligned}\phi_{acc} &= \text{atan2}\left(a_y, \sqrt{a_x^2 + a_z^2}\right) \\ \theta_{acc} &= \text{atan2}\left(-a_x, \sqrt{a_y^2 + a_z^2}\right)\end{aligned}$$

$$z_k = Cx_k + v_k, \quad v \sim N(0, R)$$

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}, z = \begin{bmatrix} \phi_{acc} \\ \theta_{acc} \end{bmatrix}, R = \begin{bmatrix} \sigma_\phi & 0 \\ 0 & \sigma_\theta \end{bmatrix}$$

Kalman Filter Equations

Predict.

$$\hat{x}^- = A \hat{x} + Bu$$

$$P^- = APA^T + Q$$

Update.

$$\tilde{y} = z - C\hat{x}^-$$

$$S = CP^-C^T + R$$

$$K = P^-C^TS^{-1}$$

$$\hat{x} = \hat{x}^- + k\tilde{y}$$

$$P = (I - KC)P^-$$

5. Extending The State Observer for Orientation Estimation (With Yaw)

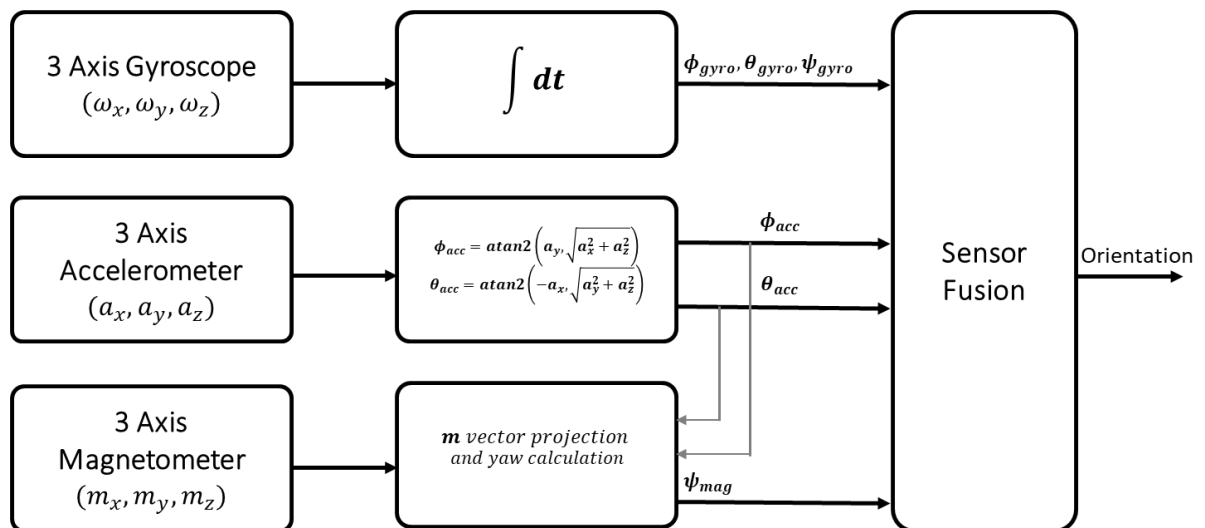


Figure 2: Overview of the System

Magnetometer

Measures the local magnetic field along each of the body frame axes (μT). The Earth's magnetic field expressed in the body frame is the fixed-frame field rotated into the body frame, corrupted by sensor distortions:

$$m^B = \begin{bmatrix} m_x \\ m_y \\ m_z \end{bmatrix} = \underbrace{A(R_E^B m_E)}_{\text{Rotated Field and Soft Iron effect}} + \underbrace{\begin{bmatrix} | \\ b_{hi} \\ | \end{bmatrix}}_{\text{Hard iron effect}} + \begin{bmatrix} | \\ \text{Noise} \\ | \end{bmatrix}$$

Where,

m_E = Earth's field in the fixed (navigation) frame

R_E^B = rotation from fixed to body frame

A = soft iron distortions/ scale factors / axis misalignments

b_{hi} = hard – iron offset (assumed to rotate with the sensor)

Calibration

Before the field can be used for heading, the hard-iron and soft-iron errors must be removed (one-time calibration). $\mathbf{b}_{hi}$ and $\mathbf{A}$ are found by rotating the sensor through all orientations and fitting an ellipsoid using convex optimization.

$$\mathbf{m}_{cal} = \begin{bmatrix} m'_x \\ m'_y \\ m'_z \end{bmatrix} = \mathbf{A}^{-1}(\mathbf{m}_{meas} - \mathbf{b}_{hi})$$

$$\hat{\mathbf{m}} = \frac{\mathbf{m}}{\|\mathbf{m}_{cal}\|} \text{ to normalize}$$

Tilt Compensation (Correction Using Roll and Pitch)

The calibrated field still lies in the tilted body frame. To extract heading it must be de-rotated into the horizontal plane using the *roll* (ϕ) and *pitch* (θ) estimates from the filter — NOT from the magnetometer:

$$\begin{aligned} m_x^h &= m'_x \cos \theta + m'_y \sin \phi \sin \theta + m'_z \cos \phi \sin \theta \\ m_y^h &= m'_y \cos \phi - m'_z \sin \phi \end{aligned}$$

Heading (Measurement)

$$\psi_{mag} = \text{atan2}(m_y^h, m_x^h) + \delta_{decl}$$

where δ_{decl} is the local magnetic declination (include only if true north is required; omit for relative heading).

Notes:

- Roll and pitch enter only as inputs to the de-rotation; the magnetometer never corrects ϕ or θ .
- The innovation/ error ψ_{mag} must be wrapped to the shortest arc in $(-\pi, \pi]$ before the update.

$$v_\psi = \psi_{mag} - \hat{\psi}^- = \text{atan2}(\sin v_\psi, \cos v_\psi)$$

- For further details of the calibration procedure including proofs and derivations, refer the attached pdf.

Euler rate kinematics

$$\begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} = \begin{bmatrix} 1 & \sin(\phi)\tan(\theta) & \cos(\phi)\tan(\theta) \\ 0 & \cos(\phi) & -\sin(\phi) \\ 0 & \frac{\sin(\phi)}{\cos(\theta)} & \frac{\cos(\phi)}{\cos(\theta)} \end{bmatrix} \begin{bmatrix} p \\ q \\ r \end{bmatrix}$$

In scalar form,

$$\begin{aligned} \dot{\psi} &= \frac{\sin(\phi)}{\cos(\theta)} q + \frac{\cos(\phi)}{\cos(\theta)} r \\ \dot{\phi} &= p + \sin(\phi)\tan(\theta) q + \cos(\phi)\tan(\theta) r \\ \dot{\theta} &= \cos(\phi) q - \sin(\phi) r \end{aligned}$$

State vector:

$$x = [\phi \quad b_\phi \quad \theta \quad b_\theta \quad \psi \quad b_\psi]^T$$

Where, $\phi = \text{roll}$, $\theta = \text{pitch}$, $\psi = \text{yaw}$ and b_ϕ, b_θ, b_ψ are bias states modeled as random-walk constants

Discrete - Time Process Model

$$\begin{aligned} \phi_k &= \phi_{k-1} + \Delta t(\dot{\phi} - b_\phi) \\ b_{\phi,k} &= b_{\phi,k-1} \\ \theta_k &= \theta_{k-1} + \Delta t(\dot{\theta} - b_\theta) \\ b_{\theta,k} &= b_{\theta,k-1} \\ \psi_k &= \psi_{k-1} + \Delta t(\dot{\psi} - b_\psi) \\ b_{\psi,k} &= b_{\psi,k-1} \end{aligned}$$

$$x_k = Ax_{k-1} + Bu_{k-1} + w_{k-1}, \quad w \sim N(0, Q)$$

$$A = \begin{bmatrix} 1 & -\Delta t & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -\Delta t & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & -\Delta t \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}, B = \begin{bmatrix} \Delta t & 0 & 0 \\ 0 & 0 & 0 \\ 0 & \Delta t & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \Delta t \\ 0 & 0 & 0 \end{bmatrix}, u = \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix}, Q = 0.01I_4$$

Measurement model

Roll and pitch are directly provided by the accelerometer readings and are treated as measurements in the state – space model.

$$\begin{aligned} \phi_{acc} &= \text{atan2}\left(a_y, \sqrt{a_x^2 + a_z^2}\right) \\ \theta_{acc} &= \text{atan2}\left(-a_x, \sqrt{a_y^2 + a_z^2}\right) \end{aligned}$$

$$m_{cal} = \begin{bmatrix} m'_x \\ m'_y \\ m'_z \end{bmatrix} = A^{-1}(m_{meas} - b_{hi})$$

$$\hat{m} = \frac{m_{cal}}{\|m_{cal}\|}$$

Tilt removal using roll and pitch

$$m_x^h = m'_x \cos \theta + m'_y \sin \phi \sin \theta + m'_z \cos \phi \sin \theta$$

$$m_y^h = m'_y \cos \phi - m'_z \sin \phi$$

Heading (Measurement)

$$\psi_{mag} = \text{atan2}(m_y^h, m_x^h) + \delta_{decl}$$

Measurement

$$z_k = Cx_k + v_k, \quad v \sim N(0, R)$$

$$z = \begin{bmatrix} \phi_{acc} \\ \theta_{acc} \\ \psi_{mag} \end{bmatrix}, C = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}, R = \begin{bmatrix} \sigma_\phi & 0 & 0 \\ 0 & \sigma_\theta & 0 \\ 0 & 0 & \sigma_\psi \end{bmatrix}$$